

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CE) Examination
May 2018

CE320/305.01/305 COMPUTER NETWORK & INTERNETWORKING LAYERS**Date: 01/05/2018, Tuesday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following:**

- (a) How many redundant bits are required to detect and correct error in the message 100011100? [01]
- (b) OSI model is known as theoretical model, why? [02]
- (c) For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology? [02]
- (d) Mention the types of error in bit during communication may occur. Find burst error length in following, Sent bit: 100011111001 and Received bit: 100111111000. [02]

Q – 2 Do as Directed [Any Three]. [12]

- (a) What is medium access control? Write working of Pure ALOHA and slotted ALOHA.
- (b) Write design issues of Data link layer. What is piggybacking? Write mechanism of sliding window protocol.
- (c) What is the use of CRC method in communication? What is the remainder obtained by dividing $X^7 + X^5 + 1$ by the generator polynomial $X^3 + 1$?
- (d) Explain hidden node and exposed node problem. Can this problem occur in wired network?

Q – 3 Do as Directed [Any Four]. [16]

- (a) What is flow control? List out protocols that can be used to synchronize data flow between sender and receiver.
- (b) Write short note on following network devices:
 - i. Switch
 - ii. Repeater
 - iii. Access point (AP)
 - iv. Router
- (c) Do comparison between fast Ethernet and gigabit Ethernet.

- (d) Explain CSMA/CA. What is RTS and CTS mechanism in wireless local area network?
- (e) Explain a simplex stop-and-wait protocol for an error-free channel
- (f) What is framing and its type? A bit string, 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing?

SECTION – II

Q - 4 Answer the following:

- (a) What is size of TCP (without any options) and UDP header in bytes? [01]
- (b) In class C IPV4 addressing, what will be subnet mask if we form 8 subnetworks? [02]
- (c) Write Size of IPV4, IPV6 and MAC address in bytes. [02]
- (d) What is the functioning of application layer? [02]

Q – 5 Do as Directed [Any Three]. [12]

- (a) Why does Count to infinity problem occur in DSV routing? How LSR routing overcome the problem of count to infinity? Write all steps of LSR routing.
- (b) List out The primitives for a simple transport service with its meaning. List out socket primitives for TCP.
- (c) Explain IS-IS and OSPF protocol. Listed protocols based on which routing algorithms: RIP, OSPF, BGP, and IS-IS.
- (d) Explain TCP and UDP protocol on the basis of connection establishment.

Q - 6 Do as Directed [Any Four]. [16]

- (a) Draw the header format of IPV4 and give briefing of each field.
- (b) Do comparison of Virtual circuit and Datagram Network over various issues.
- (c) Write short notes on Email, DNS, TFTP and DHCP protocol.
- (d) What is loopback IP address? Find network and broadcast address for given IPV4 Address 195.16.17.24/25.
- (e) What is congestion control? Explain leaky bucket and load shedding method to control congestion in network.
- (f) What is SCTP protocol? Explain the role of it in Transport Layer.
